ABSTRACT

ReliefSphere : An Intelligent Community Resource Redistribution and Decision Support Platform

Resource scarcity and unequal distribution remain significant challenges despite the availability of surplus resources within communities. Restaurants, hotels, educational institutions, companies, and individuals frequently possess reusable resources such as food, clothes, and books, while NGOs, orphanages, old-age homes, shelters, and government schools often face shortages. Existing donation platforms primarily facilitate communication between donors and recipients but offer limited support in identifying the most appropriate beneficiary organization based on real-time need.

ReliefSphere AI is a web-based community resource redistribution platform that addresses this gap through an AI-powered Multi-Criteria Decision Support System. The platform enables verified donors to contribute surplus food, clothes, and books, while verified recipient organizations receive donations or raise resource requests. Dedicated modules for donors, recipient organizations, volunteers, and administrators streamline the complete donation lifecycle — registration, verification, resource management, volunteer coordination, delivery tracking, and administrative monitoring.

The recommendation engine analyzes multiple decision factors — resource category, urgency, geographical proximity, and organizational capacity — to rank and recommend the most suitable recipient organizations for each donation. The platform further incorporates a Priority Campaign Management System that allows administrators to launch time-bound campaigns for disaster relief, educational drives, seasonal clothing collections, and other community initiatives; during an active campaign, the AI dynamically reprioritizes affected organizations to accelerate resource allocation in critical situations.

The mini project implements a fully functional resource redistribution platform integrated with a rule-based, multi-criteria AI recommendation module. The main project extends this foundation along two directions: first, by deepening the recommendation engine with Explainable AI, predictive demand analytics, trust scoring, and an adaptive weighting mechanism that learns from accepted and rejected recommendations; and second, by extending platform access to a mobile application for the Donor, Recipient Organization, and Volunteer roles, enabling field-level participation beyond the web interface.

The system is built using React.js for the frontend, Node.js with Express.js for the backend, PostgreSQL as the relational database, and Python (Scikit-learn) for the AI recommendation module, with JWT-based authentication and Leaflet/OpenStreetMap for geolocation. The use of JavaScript across both frontend and backend simplifies development and maintenance by unifying the technology stack. The primary research contribution of the project is the development and evaluation of an AI-based Multi-Criteria Decision Support Framework for community resource redistribution — one that improves transparency, efficiency, and responsiveness in humanitarian resource management, while remaining scalable enough for real-world deployment beyond the academic scope.